

# HEMAL PATEL

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## PROFESSIONAL SUMMARY

Senior Software Engineer with over 11 years of experience architecting multiplayer gameplay systems, custom engine tools, and real-time interactive pipelines across Unreal Engine and Unity. Specializes in optimization under strict hardware constraints, building server-authoritative frameworks, and leading cross-functional engineering teams to ship live-service platforms on schedule.

## TECHNICAL SKILLS

**Engines:** Unreal Engine 4/5, Unity

**Additional Engine:** Godot, Defold, Panda3D

**Languages:** C++, C#, Python, Lua

**Gameplay Engineering:** Multiplayer Networking, Server-Authoritative Architecture, Gameplay Systems, UI, Multithreading, Performance Profiling & Optimization

**Tools & Pipeline:** Editor Tools, CI/CD, BuildGraph, RenderDoc, Git, Perforce, Docker, Linux, JIRA

**Online & Backend:** Meta XR SDKs, PlayFab, NATS, REST APIs, AWS, Grafana

**AI Workflow:** Agentic Tooling and Coding Workflows, Local LLM Deployment, Context Window Optimization

## PROFESSIONAL EXPERIENCE

### Software Engineer II

*WaveXR*

**Nov 2019 – Dec 2025**

*Remote (Horsham, PA)*

- Managed a team of four engineers and technical artists to deliver the Social Venue platform that includes live gameplay frameworks. Coordinated directly with producers and product stakeholders to define scope and technical roadmaps.
- Architected a custom Unreal-to-NATS messaging layer to handle real-time performer and attendee data serialization. Built accompanying UMG editor tools that allowed technical artists to configure data routing without writing code.
- Designed a proxy system over NATS streams using a decoupled C++ MVVM pattern. This allowed runtime configuration overrides via Blueprints, isolating visual scripting from core systems and supporting cross-peripheral input switching.
- Rebuilt a high-draw Niagara particle system using the command pattern to optimize execution under full client load. Stabilized real-time performance on the hardware-constrained Quest 3, recovering frame rate from 15 fps to 36 fps with ASW enabled.
- Profiled and eliminated critical RPC bottlenecks and Blueprint overhead using Unreal Insights. Achieved an 80% reduction in frame times and optimized runtime memory allocation.
- Bypassed slow TeamCity CI/CD build loops by reverse-engineering Unreal's packaging pipeline to create a custom, UMG-based Editor Export Tool, letting content creators instantly generate local build artifacts.
- Maintained cross-platform runtime systems across Unity and UE5 for live, high-occupancy virtual concerts. Partnered with DevOps to monitor and triage client-side infrastructure anomalies using Grafana.
- Identified and resolved deterministic cross-platform bugs between Windows and Linux builds, addressing path separator issues and rendering discrepancies.
- Built a modular telemetry subsystem that batched network events before dispatch, minimizing local network overhead and optimizing data pipeline traffic.
- Integrated Meta Horizon party features and player-blocking mechanics via the Oculus SDK, keeping replication states synced across connected peers.
- Migrated legacy virtual environment codebases from Unity to UE5, porting core subsystems to native C++ to retain feature parity.

### Unity Programmer (Contract)

*Razor Edge Games*

**Nov 2015 – Nov 2019**

*Remote (Horsham, PA)*

- Engineered core runtime mechanics in C#, including modular inventory frameworks and turn-based simulation systems designed for long-term scalability.
- Built an end-to-end Character Creator system handling complex mesh deformations, skeleton animation states, and UI sub-frameworks, reducing asset implementation time for design teams.

- Developed a custom beat-detection framework and an intuitive, script-configurable audio editor tool from scratch, so designers could synchronize runtime gameplay mechanics directly to audio tracks.

#### Unity Developer

Jun 2014 – Mar 2015

Smartcloud Infotech Pvt. Ltd.

Pune, India

- Developed and shipped three real-time interactive applications across different genres, adhering strictly to engine performance targets and project deadlines.
- Provided technical estimates and structured workflow recommendations during early discovery phases to establish stable engine pipelines.
- Implemented UI layout rendering systems and core gameplay interaction loops, iterating rapidly based on user testing data to maximize runtime performance.

### ADDITIONAL EXPERIENCE

#### AI Trainer – Game Development Expert (Contract)

April 2026 – Present

Handshake AI Fellowship

Remote (Horsham, PA)

- Design and evaluate specialized technical prompts to test large language model capability in gameplay programming, systems design, and engine architecture.
- Review generated code outputs and engineering solutions for technical accuracy, performance, and alignment with game industry best practices.

#### Senior Staff Writer (Volunteer)

November 2016 – Present

Gaming Cypher

Remote (Horsham, PA)

- Write previews and reviews covering AAA and indie titles across major publishers and platforms, published at [gamingcypher.com](http://gamingcypher.com)
- Attend industry conventions including PAX East to provide on-the-ground written coverage of upcoming releases and studio announcements

### TECHNICAL PROJECTS

#### ElementalArsenal: GAS Extension Plugin

C++, Unreal Engine 5, GAS, Editor Tools

- Designed and implemented a modular plugin extending GAS to support custom equipment pipelines and automated elemental reactions, applying vector math and collision logic for spatial gameplay mechanics.
- Built custom Editor Tools to automate asset generation for GAS, accelerating the workflow for gameplay designers, using Slate and UMG.

#### NeoDataSync: Multiplayer Replication Plugin

C++, Unreal Engine, Multiplayer

- Architected an Unreal Engine plugin to streamline replication of arbitrary data structures in multiplayer environments, utilizing clean C++ serialization wrappers to handle complex gameplay states.

#### NeoThreads: Unreal Multithreading Framework

C++, Unreal Engine, Multithreading

- Developed a lightweight concurrency framework to safely offload heavy calculations from the Game Thread, improving frame stability and performance for CPU-intensive gameplay systems.

### SHIPPED

Wave VR (Unity) | Wave Beta (Unreal) | Wave Standalone (Unreal) | Alimentary Avengers (Unity) | Tafe Planetary Plus (Unity) | Be Safe, Be Smart!

Notable Experiences: The Weeknd | Justin Bieber | Calvin Harris | John Legend | Tinashe | Pentakill

### EDUCATION

#### DePaul University

Master of Science in Computer Game Programming (Coursework Completed)

#### CEPT University

Master of Business Administration, Information Technology & Project Management

#### Gujarat Technological University

Bachelor of Technology in Information Technology